universität wien
Faculty of Psychology

Lernen unter COVID-19-Bedingungen

Pupils

Method Report

1. Introduction

The present cross-sectional study is the second phase of the project "Learning under COVID-19 conditions" with pupils previously archived at https://doi.org/10.11587/VRQL3B.

The first phase contained three cross-sectional measurement points. The second phase of the project comprises two additional cross-sectional measurement points, labelled as measurement point 4 (MZP4) and 5 (MZP5). As in the previous phase, the project aims at investigating learning process under COVID-19 conditions, extending its focus to pupils' behaviour and coping in school in time of pandemic. These two additional measurement points again include data from large samples of Austrian secondary school pupils, collected in November/December 2020 and then in April/May 2021.

2. Study Description

2.1. Title and version

Lernen unter COVID-19-Bedingungen: SchülerInnen (Learning under COVID-19 conditions: pupils, V2.0)

2.2. Principal Investigators

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2.3. Funding/Acknowledgments

The present study has been funded by the Vienna Science and Technology Fund (WWTF), by MEGA Bildungsstiftung, and the City of Vienna through project COV20-025.

2.4. Fieldwork agency

The fieldwork of the study "Lernen unter COVID-19-Bedingungen" was managed and handled internally.

2.5. Topics/Keywords

- Sociodemographic information: age, sex, education, language, region
- Situation at home: household structure, help provided, accessible electronic devices, socio-economic status indicators
- Crisis perception: school attendance, feelings of safety, information, perceived importance and behavioral adherence to socially responsible behavior

- Effects of the crisis: changes in well-being, anxiety and depression, learning, social relatedness, competence, autonomy, and self-regulated learning
- Learning: basic needs (social relatedness, competence, autonomy), self-regulated learning, learning goals, learning time, learning environment, metacognition, procrastination
- Education: achievement, school climate
- Online-education: digital learning, cyber bulling

3. Study Design

3.1. Sampling

The sample of the study "Lernen unter COVID-19-Bedingungen" consisted of Austrian pupils attending secondary school with access to the internet. The pupils were invited to participate in the survey by their teachers, headmasters, and by calls in radio and television programs. Moreover, the link to and a recommendation to participate in the survey was published by the Federal Ministry of Education, Science, and Research on their website. The link with a respective invitation to the study was also placed on the survey's website. In order to participate in the study, the respondents had to agree to the informed consent, be Austrian residents, and attend secondary school. It is important to note that the sample cannot be considered representative for the Austrian population as many students do not have easy access to the internet and/or devices needed to fill out the survey. Therefore, risk groups may be underrepresented in the current sample.

3.2. Fieldwork

The overall data collection was conducted in the time from December 2020 to May 2021.

- Wave 1 (MZP4): 22 November 2020 – 06 December 2020
- Wave 2 (MZP5): 06 April 2021 – 25 May 2021

4. Data Cleaning

4.1. General data cleaning procedure

Cleaning procedure was the same for data sets collected in both measurement points and contained two steps:

4.1.1. Initial data cleaning

As participating students answered an online questionnaire via Unipark (Questback, 2007), data is exported from unipark in .csv format and imported to SPSS. Variables names are replaced. All missing values are defined as -99.

Invalid zero values or values of -77 are replaced with -99. Participants who gave consent, and had answering time (variable duration) greater than zero are selected for a new SPSS file.

4.1.2. Data cleaning

As unipark assigns -1 to variable duration in cases where the break was made during the answering the questionnaire, all valid -1 cases are replaced with (+) 9999999 value. Participants with 100% missings on main outcome variables (positive affect, social relatedness, competence, autonomy, SRL scales and procrastrination in the fourth measurement point; and positive affect and self-efficacy in the fifth measurement point) are excluded. Minimum possible duration is tested by filling in the questionnaire as fast as possible while reading all items. All cases above the duration treshold and within a define age-range (based on data from Statistics Austria) are selected for a new SPSS file.

4.1.3. Missing values

All empty spaces are replaced with -99. Missing values that are represented by 0 in the raw data set are replaced by -99 on all variables where 0 is not a valid value. Missing values represented with -77 in unipark are replaced with -99 as well. Missing values on variables that resulted from the application of filters are defined with -98. Cases where participants by mistake got to see a question they should not have seen (due to technical issues in unipark) are replaced with -78.

All values above 24 on variable dp.school.curr.contact are set to missing (-99).

4.1.4. Filters

Data records of participants who did not give their consent are deleted.

A fastest possible time for answering the questionnaire is determined diligently. Variable duration of answers with duration below the fastest possible time is set to 9999999.

Data records with 100% missings on main outcome variables are deleted.

All other filters are defined in the questionnaire, as well as in data set using values -98 and -78 (see above) as value labels.

4.1.5. Answers to open questions

All open questions are removed from the data sets due to anonymization rules.

4.1.6. Multiple choice / single choice

“Welche Sprache(n) sprichst du zu Hause?” (dp.lang.ger, dp.lang.eng, dp.lang.tur, dp.lang.bcs, dp.lang.hun, dp.lang.slo, dp.lang.pol, dp.lang.arb, dp.lang.per, dp.lang.rus, dp.lang.other) and „Falls du derzeit zu Hause Hilfe beim Lernen bekommst, wer hilft dir vor allem?“ (dp.mother, dp.father, dp.siblings, dp.other) are multiple choice questions in MZP4.

“Welche Dinge gibt es bei dir zuhause?” (ek.ses1.1, …, ek.ses1.5) is a multiple choice question in MZP5.

4.2. Anonymisation procedure

4.2.1. Code variables

Due to the high risk for identification, variables ek.code (personal code) and ek.class.code (identifier for matching pupils of the same class) were deleted from the data set.

4.2.2. Language variables

Data on the languages spoken were gathered via multiple choice and an open answer box for additional languages that were not eligible in the multiple choice question. The information from the open answers (dp.lang.other) is reduced to a dichotomous variable (“yes”/“no”) that indicates if additional languages were named. Answers indicating “no”, unclear inputs and missing answers are set as “no”.

In MZP5, the selected display languages other than German (ek.language) are categorized to get groups with at least 20 observations. For ek.ger.class.compr, answers “eher schlecht” and “schlecht“ are merged to the category “eher schlecht oder schlecht“.

4.2.3. Gender variables

In MZP5, answers indicating that the corresponding person chose “divers” (ek.gender) are recoded to missing as there are less than 20 observations.

4.2.4. Age variables

Data containing the age of the participants in MZP5 are categorized by grouping outliers to get groups with at least 20 observations.

4.2.5. School type variables

In the school type variable in MZP5 (ek.school), school types that occur rarely are categorized to “andere” to get groups with at least 20 observations.

4.2.6. Federal state variables

In MZP5, for ek.school.fs, answers “Burgenland” and “Niederösterreich” are merged to “Burgenland/Niederösterreich”, answers “Tirol” and “Vorarlberg” are merged to “Tirol/Vorarlberg”.

4.2.7. Closed questions

The following answer categories were merged in MZP4 to get groups with at least 20 observations:

Variable(s)	Answers merged	New value label

dp.lang.learn	all numerical values indicating 19 or 20	19 oder 20
dp.kids	all numerical values indicating 6 or more	6 oder mehr Kinder
dp.school.curr.contact	all numerical values indicating 11 or more	11 oder mehr

The following answer categories were merged in MZP5 to get groups with at least 20 observations:

Variable(s)	Answers merged	New value label
ek.noattend.school.why	• weil ich zur Risikogruppe gehöre. • weil jemand in meinem Umfeld/meiner Familie zur Risikogruppe gehört. • weil derzeit meine Klasse aufgrund eines positiven Corona-Falls in Distance Learning ist. • weil derzeit meine Schule aufgrund mehrerer positiver Corona-Fälle in Distance Learning ist. • Ich weiß es nicht. • Ich möchte es nicht sagen.	Anderes
ek.persons	all numerical values indicating 7 or more	7 oder mehr
ek.kids	all numerical values indicating 4 or more	4 oder mehr
ek.adults	all numerical values indicating 5 or more	5 oder mehr
ek.employ	all numerical values indicating 4 or more	4 oder mehr
ek.norms.imp3	• eher nicht wichtig • gar nicht wichtig	eher nicht wichtig oder gar nicht wichtig
ek.norms.adh1 ek.norms.adh3	• eher nicht • gar nicht	eher nicht oder gar nicht
ek.grade.german	• 4-5 • 5	4-5 oder 5
ek.cyberb2 ek.cyberb4 ek.cyberb6	• ungefähr einmal pro Woche • fast jeden Tag	ungefähr einmal pro Woche oder öfter
ek.cyberb3 ek.cyberb5 ek.cyberb7 ek.cyberb8	• zwei- oder dreimal im Monat • ungefähr einmal pro Woche • fast jeden Tag	zwei- oder dreimal im Monat oder öfter

4.2.8. Open questions

In MZP4, ten open variables are deleted without recoding or standardizing the answers:

- dp.who ("Falls du derzeit zu Hause Hilfe beim Lernen bekommst, wer hilft dir vor allem? Andere")
- dp.platform ("Wenn ihr für die Schule derzeit Lernplattformen benutzt: Welche sind das?")
- dp.com.platform ("Welche Kommunikationsmedien (z.B. WhatsApp, Twitter, ...) nutzt du, um mit deinen Lehrer/innen in Kontakt zu bleiben?")
- dp.info.more ("Worüber würdest du mehr Information brauchen?")
- dp.studya.better ("Wenn sich etwas verbessert hat: Was ist das genau?")
- dp.studya.worse ("Wenn sich etwas verschlechtert hat: Was ist das genau?")
- dp.future ("Wenn du an die nächsten Wochen denkst, worüber machst du dir am meisten Gedanken?")

In MZP5, five open variables are deleted without recoding or standardizing the answers:

- ek.noattend.school.why.other ("Warum gehst du derzeit nicht in die Schule? Anderes")

- ek.school.other ("Welche Schulart besuchst du? Andere")
- ek.help.other.who ("Bekommst du derzeit genug Unterstützung beim Lernen? Andere")
- ek.challenges ("Was waren für dich seit Beginn der Pandemie die größten Herausforderungen?")
- ek.accomplish ("Was ist dir seit Beginn der Pandemie besonders gut gelungen?")

4.2.9. Open questions containing numeric values

Answers to questions requiring numeric values are recoded according to the description in section "Data cleaning". Original (string) answers are deleted.

4.3. Further deletions

4.3.1. Consent variable

One variable containing the participants' agreement on processing the filled in data for scientific purposes is deleted in MZP4 (dp.consent) and MZP5 (ek.consent) as all included participants gave their consent for processing the filled in data for scientific purposes.

4.3.2. Filter variables

One filter variable (filter_$) originating from the cleaning procedure is deleted in each measurement point.

4.3.3. School code variable

One variable (ek.school.code) that contains a code identifying the school that the student is attending is deleted in MZP5.

4.3.4. Variables with few valid values

In MZP5, one variable that contains only missing values (quality) and one variable that contains only two valid values (ek.bele.grade) are deleted.

4.3.5. System variables

System variables that contain information of the duration of answering separate pages are deleted.

MZP4:

- date_of_first_mail (date of first e-mail)
- datetime (date of first access)
- date_of_last_access (date of last access)
- v_357 (captcha question)
- v_358 (expected answer to captcha question)

- captcha_answer (answer to captcha question)
- captcha_answer_numeric (numeric answer to captcha question)
- rts4787304, rts4787307, rts4787309, rts4787311, rts4787312, rts4787313, rts4787314, rts4787315, rts4787316, rts4787317, rts4787318, rts4787319, rts4787323, rts4787324, rts4792323, rts4792362, rts4792422, rts4792480, rts4792747, rts4792770, rts4794861, rts4795386, rts4795404 (relative time stamps for pages)

MZP5:

- external_lfdn (external sequence number)
- tester (filter variable indicating entries made for test purposes)
- dispcode (disposition code)
- lastpage (identifier of exit page)
- browser (browser identifier)
- referer (HTTP referer)
- page_history (array of page identifiers indicating the sequence of visited questionnaire pages)
- hflip (questionnaire answered on a screen in landscape mode)
- vflip (questionnaire answered on a screen in portrait mode)
- session_id (session identifier)
- ats (absolute timestamp)
- datetime (date of first access)
- date_of_last_access (date of last access)
- date_of_first_mail (date of first e-mail)
  rts4909219, rts4909223, rts4909226, rts4909231, rts4909232, rts4909233, rts4909234, rts4909240, rts4909242, rts4909244, rts4909292, rts4909299, rts4909323, rts4909398, rts4909401, rts4909407, rts4909588, rts4909613, rts4909657, rts4919653, rts4959324, rts4985049, rts4991305, rts4992501, rts5008303, rts5008309 (relative time stamps for pages)

4.3.6. Reverse-coded variables

All reverse-coded variables are deleted.

MZP4:

- dp.teacher.support.rec (Reversed Likert scale of item dp.teacher.support)
- dp.media.rec (Reversed Likert scale of item dp.media)
- dp.media.tax.rec (Reversed Likert scale of item dp.media.tax)
- dp.school.hw.rec (Reversed Likert scale of item dp.school.hw)
- dp.sn.learn1.rec (Reversed Likert scale of item dp.sn.learn1)
- dp.sn.learn2.rec (Reversed Likert scale of item dp.sn.learn2)
- dp.sn.learn3.rec (Reversed Likert scale of item dp.sn.learn3)

- dp.sn.learn4.rec (Reversed Likert scale of item dp.sn.learn4)
- dp.sn.learn5.rec (Reversed Likert scale of item dp.sn.learn5)
- dp.sn.cor1.rec (Reversed Likert scale of item dp.sn.cor1)
- dp.sn.cor2.rec (Reversed Likert scale of item dp.sn.cor2)
- dp.sn.cor3.rec (Reversed Likert scale of item dp.sn.cor3)
- dp.sn.cor4.rec (Reversed Likert scale of item dp.sn.cor4)
- dp.sn.cor5.rec (Reversed Likert scale of item dp.sn.cor5)
- dp.sn.cor6.rec (Reversed Likert scale of item dp.sn.cor6)
- dp.info.rec (Reversed Likert scale of item dp.info)
- dp.achiev.rec (Reversed Likert scale of item dp.achiev)
- dp.achiev.satis.rec (Reversed Likert scale of item dp.achiev.satis)
- dp.pa1.rec (Reversed Likert scale of item dp.pa1)
- dp.pa2.rec (Reversed Likert scale of item dp.pa2)
- dp.pa3.rec (Reversed Likert scale of item dp.pa3)
- dp.pa.change.rec (Reversed Likert scale of item dp.pa.change)
- dp.study.curr.rec (Reversed Likert scale of item dp.study.curr)
- dp.studya.change.rec (Reversed Likert scale of item dp.studya.change)
- dp.lm1.rec (Reversed Likert scale of item dp.lm1)
- dp.lm2.rec (Reversed Likert scale of item dp.lm2)
- dp.lm3.rec (Reversed Likert scale of item dp.lm3)
- dp.lm.change.rec (Reversed Likert scale of item dp.lm.change)
- dp.sr1.rec (Reversed Likert scale of item dp.sr1)
- dp.sr2.rec (Reversed Likert scale of item dp.sr2)
- dp.sr3.rec (Reversed Likert scale of item dp.sr3)
- dp.sr.change.rec (Reversed Likert scale of item dp.sr.change)
- dp.comp1.rec (Reversed Likert scale of item dp.comp1)
- dp.comp2.rec (Reversed Likert scale of item dp.comp2)
- dp.comp3.rec (Reversed Likert scale of item dp.comp3)
- dp.comp.change.rec (Reversed Likert scale of item dp.comp.change)
- dp.auto1.rec (Reversed Likert scale of item dp.auto1)
- dp.auto2.rec (Reversed Likert scale of item dp.auto2)
- dp.auto3.rec (Reversed Likert scale of item dp.auto3)
- dp.auto.change.rec (Reversed Likert scale of item dp.auto.change)
- dp.gp1.rec (Reversed Likert scale of item dp.gp1)
- dp.time2.rec (Reversed Likert scale of item dp.time2)
- dp.meta1.rec (Reversed Likert scale of item dp.meta1)
- dp.meta2.rec (Reversed Likert scale of item dp.meta2)
- dp.meta3.rec (Reversed Likert scale of item dp.meta3)

- dp.srl.change.rec (Reversed Likert scale of item dp.srl.change)
- dp.f1.rec (Reversed Likert scale of item dp.f1)
- dp.f2.rec (Reversed Likert scale of item dp.f2)
- dp.f3.rec (Reversed Likert scale of item dp.f3)
- dp.f4.rec (Reversed Likert scale of item dp.f4)
- dp.f5.rec (Reversed Likert scale of item dp.f5)
- dp.m1.rec (Reversed Likert scale of item dp.m1)
- dp.m2.rec (Reversed Likert scale of item dp.m2)
- dp.m3.rec (Reversed Likert scale of item dp.m3)
- dp.m4.rec (Reversed Likert scale of item dp.m4)
- dp.m5.rec (Reversed Likert scale of item dp.m5)

MZP5:

- ek.ger.frequency.rec (Reversed Likert scale of item ek.ger.frequency)
- ek.ger.class.compr.rec (Reversed Likert scale of item ek.ger.class.compr)
- ek.ses2.rec (Reversed Likert scale of item ek.ses2)
- ek.ses3.rec (Reversed Likert scale of item ek.ses3)
- ek.ses4.rec (Reversed Likert scale of item ek.ses4)
- ek.ses5.rec (Reversed Likert scale of item ek.ses5)
- ek.ses6.rec (Reversed Likert scale of item ek.ses6)
- ek.sr1.rec (Reversed Likert scale of item ek.sr1)
- ek.sr2.rec (Reversed Likert scale of item ek.sr2)
- ek.sr3.rec (Reversed Likert scale of item ek.sr3)
- ek.comp1.rec (Reversed Likert scale of item ek.comp1)
- ek.comp2.rec (Reversed Likert scale of item ek.comp2)
- ek.comp3.rec (Reversed Likert scale of item ek.comp3)
- ek.auto1.rec (Reversed Likert scale of item ek.auto1)
- ek.auto2.rec (Reversed Likert scale of item ek.auto2)
- ek.auto3.rec (Reversed Likert scale of item ek.auto3)
- ek.eff1.rec (Reversed Likert scale of item ek.eff1)
- ek.eff2.rec (Reversed Likert scale of item ek.eff2)
- ek.eff3.rec (Reversed Likert scale of item ek.eff3)
- ek.norms.imp1.rec (Reversed Likert scale of item ek.norms.imp1)
- ek.norms.imp2.rec (Reversed Likert scale of item ek.norms.imp2)
- ek.norms.imp3.rec (Reversed Likert scale of item ek.norms.imp3)
- ek.norms.adh1.rec (Reversed Likert scale of item ek.norms.adh1)
- ek.norms.adh2.rec (Reversed Likert scale of item ek.norms.adh2)
- ek.norms.adh3.rec (Reversed Likert scale of item ek.norms.adh3)

- ek.help.rec (Reversed Likert scale of item ek.help)
- ek.help.teacher.rec (Reversed Likert scale of item ek.help.teacher)
- ek.help.mother.rec (Reversed Likert scale of item ek.help.mother)
- ek.help.father.rec (Reversed Likert scale of item ek.help.father)
- ek.help.grandparents.rec (Reversed Likert scale of item ek.help.grandparents)
- ek.help.siblings.rec (Reversed Likert scale of item ek.help.siblings)
- ek.help.other.rec (Reversed Likert scale of item ek.help.other)
- ek.school.safety1.rec (Reversed Likert scale of item ek.school.safety1)
- ek.school.safety2.rec (Reversed Likert scale of item ek.school.safety2)
- ek.school.safety3.rec (Reversed Likert scale of item ek.school.safety3)
- ek.school.safety4.rec (Reversed Likert scale of item ek.school.safety4)
- ek.school.info1.rec (Reversed Likert scale of item ek.school.info1)
- ek.school.info2.rec (Reversed Likert scale of item ek.school.info2)
- ek.school.info3.rec (Reversed Likert scale of item ek.school.info3)
- ek.school.info4.rec (Reversed Likert scale of item ek.school.info4)
- ek.school.outcome1.rec (Reversed Likert scale of item ek.school.outcome1)
- ek.school.outcome2.rec (Reversed Likert scale of item ek.school.outcome2)
- ek.school.outcome3.rec (Reversed Likert scale of item ek.school.outcome3)
- ek.school.outcome4.rec (Reversed Likert scale of item ek.school.outcome4)
- ek.school.emot1.rec (Reversed Likert scale of item ek.school.emot1)
- ek.school.emot2.rec (Reversed Likert scale of item ek.school.emot2)
- ek.school.emot3.rec (Reversed Likert scale of item ek.school.emot3)
- ek.school.emot4.rec (Reversed Likert scale of item ek.school.emot4)
- ek.school.emot5.rec (Reversed Likert scale of item ek.school.emot5)
- ek.school.emot6.rec (Reversed Likert scale of item ek.school.emot6)
- ek.depr1.rec (Reversed Likert scale of item ek.depr1)
- ek.depr2.rec (Reversed Likert scale of item ek.depr2)
- ek.depr3.rec (Reversed Likert scale of item ek.depr3)
- ek.depr4.rec (Reversed Likert scale of item ek.depr4)
- ek.anx1.rec (Reversed Likert scale of item ek.anx1)
- ek.anx2.rec (Reversed Likert scale of item ek.anx2)
- ek.anx3.rec (Reversed Likert scale of item ek.anx3)
- ek.anx4.rec (Reversed Likert scale of item ek.anx4)
- ek.gp1.rec (Reversed Likert scale of item ek.gp1)
- ek.time2.rec (Reversed Likert scale of item ek.time2)
- ek.meta1.rec (Reversed Likert scale of item ek.meta1)
- ek.meta2.rec (Reversed Likert scale of item ek.meta2)
- ek.meta3.rec (Reversed Likert scale of item ek.meta3)

- ek.emotcoping1.rec (Reversed Likert scale of item ek.emotcoping1)
- ek.emotcoping2.rec (Reversed Likert scale of item ek.emotcoping2)
- ek.emotcoping3.rec (Reversed Likert scale of item ek.emotcoping3)
- ek.emotcoping4.rec (Reversed Likert scale of item ek.emotcoping4)
- ek.problcoping1.rec (Reversed Likert scale of item ek.problcoping1)
- ek.problcoping2.rec (Reversed Likert scale of item ek.problcoping2)
- ek.problcoping3.rec (Reversed Likert scale of item ek.problcoping3)
- ek.problcoping4.rec (Reversed Likert scale of item ek.problcoping4)
- ek.optimism1.rec (Reversed Likert scale of item ek.optimism1)
- ek.optimism2.rec (Reversed Likert scale of item ek.optimism2)
- ek.optimism3.rec (Reversed Likert scale of item ek.optimism3)
- ek.optimism4.rec (Reversed Likert scale of item ek.optimism4)
- ek.performance.rec (Reversed Likert scale of item ek.performance)
- ek.performance.satis.rec (Reversed Likert scale of item ek.performance.satis)
- ek.impulsive1.rec (Reversed Likert scale of item ek.impulsive1)
- ek.impulsive2.rec (Reversed Likert scale of item ek.impulsive2)
- ek.impulsive3.rec (Reversed Likert scale of item ek.impulsive3)
- ek.impulsive4.rec (Reversed Likert scale of item ek.impulsive4)

All variables that contain calculated mean values for each scale are deleted.

MZP4:

- dp.pa (Mean scale Positive affect (reversed items used))
- dp.lm (Mean scale Learning motivation (reversed items used))
- dp.sr (Mean scale Social relatedness (reversed items used))
- dp.competen (Mean scale Competence (reversed items used))
- dp.auto (Mean scale Autonomy (reversed items used))
- dp.meta (Mean scale Metacognition (reversed items used))
- dp.agency (Mean scale Agency (reversed items used))
- dp.communion (Mean scale Communion (reversed items used))

MZP5:

- ek.sr (Mean scale Social relatedness (reversed items used))
- ek.competen (Mean scale Competence (reversed items used))
- ek.auto (Mean scale Autonomy (reversed items used))
- ek.eff (Mean scale Efficacy (reversed items used))
- ek.safety (Mean scale School safety (reversed items used))
- ek.info (Mean scale School info (reversed items used))

- ek.outcome (Mean scale School outcome (reversed items used))
- ek.school.emotions (Mean scale School emotions (reversed items used))
- ek.pa (Mean scale Positive affect)
- ek.depr (Mean scale Depression ICD-10 (reversed items used))
- ek.anx (Mean scale Anxiety ICD-10 (reversed items used))
- ek.cyberb (Mean scale Cyberbullying)
- ek.meta (Mean scale Metacognition (reversed items used))
- ek.school.lc (Mean scale Learning climate)
- ek.emotcoping (Mean scale Emotinal coping (reversed items used))
- ek.problcoping (Mean scale Problem coping (reversed items used))
- ek.optimism (Mean scale Optimism (reversed items used))
- ek.impulsive (Mean scale Impulsivity (reversed items used))